

Wei-Jhih Huang

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Research Interests

AI-Native Radio Access Networks, Learning-Based Resource Allocation, Deep Learning for Cross-Domain Applications

Education

Bachelor, Communication Engineering, National Central University
GPA: 3.94/4.3, Expected Graduation: July 2026

Aug 2022 – Present
Taoyuan, Taiwan

Research Experience

Wireless Mobile Network Lab, National Taiwan University
Research Assistant, Advisor: **Prof. Hung-Yu Wei**

Apr 2026 – Present
Taipei, Taiwan

- Proposed CLKVO, an ***O(1)*** online policy for cross-layer KV cache orchestration in edge LLM serving, jointly optimizing handover signaling, backhaul transport, and GPU memory admission; achieved **3.3x** delay reduction over state-of-the-art.
- Developing a Lyapunov-constrained DRL extension supporting mixed RAN-AI workloads with predictive RSRP-based cache pre-staging and building trace-driven simulation pipelines for multi-BS AI-RAN evaluation.

Information Processing and Communications Lab, National Central University
Undergraduate Research Student, Advisor: **Prof. Chih-Wei Huang**

Feb 2024 – Present
Taoyuan, Taiwan

- Proposed a self-aware Meta-RL framework for 6G radio resource management with a Feature-wise Attention Gate enabling KNN-based OOD detection of traffic shifts; achieved **0.790** AUROC and **1.51 ms** latency on 100 3GPP 5QI-compliant scenarios.
- Built a trajectory-embedding mechanism for scenario distribution-aware DRL fine-tuning, reducing unnecessary DRL retraining by **90%** and improving user quality of service by **11%** in cross-layer resource management scenarios.

Electronic Design Automation Lab, The Chinese University of Hong Kong
Summer Visiting Research Student, Advisor: **Prof. Bei Yu**

Jul – Aug 2025
Hong Kong

- Enhanced **NUA-Timer++** for graph-based hierarchical circuit timing prediction under Non-Uniform Input Arrival Times, achieving a **1.96x** improvement in consistency and reducing average error to **0.42x** compared to SOTA.
- Collaborated with a postdoctoral researcher on Transformer architecture design and loss function refinement for **NUA-Timer++**; independently investigated a critical-path-based timing constraint extension as an auxiliary signal for logic synthesis.

Publications

- **Wei-Jhih Huang**, Jin-Min Yang, Ning Ma, Sheng-Liang Wu, Chih-Wei Huang. "Deviation-Aware Trajectory Embedding for Fine-Tuning Triggering in DRL-Based Wireless Network Resource Management," *In Proceedings of the IEEE Wireless Communications and Networking Conference (WCNC)*, 2026.
- **Wei-Jhih Huang**, Jin-Min Yang, Chih-Wei Huang. "Distributional Awareness for Self-Adaptive Radio Resource Management via Attention-Enhanced Meta-RL in 6G Networks," *IEEE Transactions on Network and Service Management (TNSM)*, Under Review.
- Ching-Wei Lin, **Wei-Jhih Huang**, Sheng-Liang Wu, Chih-Wei Huang. "PACI-Net: Physics-Aware Distillation for Lightweight and Explainable 6G Radio Resource Management," *IEEE Global Communications Conference (GLOBECOM)*, Under Review.
- Ziyi Wang, **Wei-Jhih Huang**, Fangzhou Liu, Tsung-Yi Ho, David Z. Pan, Bei Yu. "NUA-Timer++: Timing Prediction for Hierarchical RTL under Non-Uniform Input Arrival Times," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, Under Review.

Honors & Awards

- **Outstanding Student Scholarship**, National Central University, 2026
- **Outstanding College Youth Award**, School Level, Taiwan, 2026
- **Undergraduate Research Fellowship**, National Science and Technology Council (NSTC), Taiwan, 2025–2026
- **Honorable Mention**, Undergraduate Project Competition, Department of Communication Engineering, NCU, 2025
- **Merit Award**, Individual Service-Learning Excellence Scholarship, National Central University, 2025
- **High Distinction and Public Service Scholarship**, National Central University (as President, NCU Rural Service Club), 2024

Teaching & Work Experience

Chunghwa Telecom

AI Software Development Intern

Sep – Dec 2025

Taipei, Taiwan

- Spearheaded the full-stack development of an intelligent project management system, spanning UI/UX design and n8n workflow backend orchestration, driven by RAG-based agentic models to enhance operational efficiency by 33%.

National Central University

Teaching Assistant, Courses: *Natural Language Processing* and *Enterprise Application of Pattern Recognition*

Feb – Jun 2025

- Assisted in lecture preparation and grading for NLP topics for 60 students and image recognition topics for 60 students, analyzing the underlying AI technologies of real-world company products to support students in final project research.

Industrial Technology Research Institute

AI Software Summer Intern

Jun – Sep 2024

Hsinchu, Taiwan

- Built and deployed an LLM-based service chatbot at *Taipei Songshan Sports Center*, integrating Azure OpenAI with Linux backend services to provide 24/7 automated customer support, reducing customer queue time by 50% and improving service efficiency.